

ASSIGNMENT BRIEF

HTU Course No: 10203361	HTU Course Name: Forensics
BTEC Unit Code: T/618/7444	BTEC UNIT Name: Forensics



Student Name/ID Number/Section	
HTU Course Number and Title	10203361 Forensics
BTEC Unit Code and Title	T/618/7444 Forensics
Academic Year	2025-2026 2
Assignment Author	Safaa Hriez
Course Tutor	Safaa Hriez
Assignment Title	Data Leak
Assignment Ref No	1
Issue Date	11/04/2026
Formative Assessment dates	From 11/04/2026 to 04/06/2026
Submission Date	06/06/2026
IV Name & Date	Elham Derbas 10/04/2026
Submission Format	
The submission for this assignment is: 1. An individual written detailed forensic report showing all the steps taken in your investigation. 2. An individual written detailed report that answered the questions illustrated in the Assignment Brief and Guidance section. An oral discussion or its equivalent will be conducted after the submission of the reports. Instructions, date, and time for the discussion will be provided later. A witness statement or observation record is considered as evidence for this part. Report guidelines: In each report, you should make use of headings, paragraphs, and subsections as appropriate. The expected word limit is about 5000 word, although you will not be penalised for exceeding the total word limit. Do your best to be within the word limit. Your report should be: 1. In a form of a word file submitted via the elearning. 2. Written in a formal business style using Times New Roman and font size 12. You are required to submit the declaration form to confirm that the submitted work is entirely your own and that you fully understand the consequences of plagiarism.	
Unit Learning Outcomes	
LO1 Examine the processes and procedures for carrying out digital forensic investigation LO2 Discuss the legal and professional guidelines and procedures for carrying out digital forensic investigation LO3 Use a tool or tools to conduct digital forensic investigation on devices or networks or cyber attacks LO4 Develop a forensic examination plan and make some recommendations for use in digital forensic investigation.	
Assignment Brief and Guidance	

A confidential information leak has been discovered within a mid-sized legal consulting firm in UK. Several sensitive client contracts were found outside the organization, raising concerns that internal data had been copied and shared without authorization. Due to the legal and reputational risks, a digital forensics investigation was initiated to determine the source of the leak.

During the internal review, two employees were identified as primary suspects based on their access to the affected documents and unusual activity on their workstations.

The first suspect, Michael, is a senior legal assistant who had direct access to confidential contract files. It was observed that he had recently opened several sensitive client contracts outside of normal working hours.

The second suspect, Sarah, works as an administrative coordinator. During a routine inspection of her workstation, it was discovered that she had recently used external storage devices, despite strict company policies prohibiting such actions.

Both employees denied any involvement in the data leak. To preserve the integrity of the potential evidence, the digital forensics team created forensic images of Michael's computer and Sarah's USB.

Use this image for the laptop and this image for the USB flash drive. Your task is to analyze both pieces of evidence to identify any relevant information. Your work must be documented in a forensic report that details all the steps taken to uncover the evidence. Each step should be clearly explained and reproducible by another investigator.

Also, you have to write another report containing your answers for the following questions based on your work in the investigation:

1. Define the term of Digital Forensics and list four of its main type with brief explanation for them.
2. Did you use Rodney McKemmish model in your investigation? If yes, explain it. If no, illustrate the used model and explain its steps.
3. What is the importance of following the model presented in the previous question?
4. Evaluate the advantages and disadvantages of conducting digital forensic investigation to improve system security.
5. Given that this crime took place in the UK, name one relevant law and explain its key implications in a forensic investigation.
6. Discuss the legal and ethical requirements that you have considered when conducting the investigation.
7. The suspects in the case are citizen with legal rights. How did you ensure your investigation was ethically conducted in reference to their rights?
8. Evaluate the impact of both following and not following guidelines in a legal case, with regard to digital forensic evidence.
9. What was your forensic analysis plan before beginning the investigation? Specify the artifact(s) you planned to analyze, the tools you selected and why, and what you expected to discover.
10. Since you were not responsible for on-site evidence acquisition, list five recommendations for proper evidence acquisition in a digital forensic investigation.
11. Compare your recommendations for evidence acquisition with the [ISO/IEC 27037 guidelines](#). Focus only on the acquisition step.

Your recommendations	ISO/IEC27037 guidelines	Discussion

12. Critically evaluate your investigation, suggesting improvements to the current digital forensic investigation guidelines, processes and procedures. Your evaluation should include what did you do and why? were the tools appropriate? what aspects went well and should be repeated? what weaknesses were identified and how will you improve them in future investigations?
13. Compare the functionalities of the [Hasher](#) and Hash My files tools in hashing the files. Discuss their differences in terms of ease of use, hashing algorithms, output and export formats. Include screenshots where possible.

	Hasher	Hash My Files
Ease of use		
Hashing algorithms		
Output and export format		

Learning Outcomes and Assessment Criteria			
Learning Outcome	Pass	Merit	Distinction
LO1 Examine the processes and procedures for carrying out digital forensic investigation	P1 Discuss what is meant by digital forensics with the aid of diagrams/pictures. P2 Examine the processes and procedures for conducting digital Forensic investigation.	M1 Assess the importance of following a process or procedure when conducting digital forensic investigation	D1 Evaluate the advantages and disadvantages of conducting digital forensic investigation to improve system security
LO2 Discuss the legal and professional guidelines and procedures for carrying out digital forensic investigation	P3 3 Examine law enforcement guidelines for conducting digital forensic investigations. P4 Discuss legal and ethical requirements for conducting digital forensic investigations.	M2 Assess how ethical it is to conduct digital forensic investigation on a suspected individual, with reference to their legal rights.	D2 Evaluate the impact of both following and not following guidelines in a legal case, with regard to digital forensic evidence.
LO3 Use a tool or tools to conduct digital forensic investigation on devices or networks or cyber attacks	P5 Determine hardware and software tools that can be used to conduct digital forensic investigation. P6 Examine the file system structure of several operating systems	M3 Compare two tools that can be used to conduct digital forensic investigation. M4 Conduct a digital forensic investigation on a device or network or cyber attack.	D3 Critically evaluate the forensic investigation carried out, suggesting improvements to the current digital forensic investigation guidelines, processes and procedures
LO4 Develop a forensic examination plan and make some recommendations for use in digital forensic investigation.	P7 Develop a forensic examination plan for conducting an analysis on digital devices or networks or cyber attacks. P8 Recommend improvements based on investigation analysis	M5 Compare the recommendations for best practices for conducting digital forensics	